

Differentiation

Bring the power down, take one off. The derivative is the gradient: zero at turning points; for motion, $s \rightarrow v \rightarrow a$.



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THE TOPIC ON ONE MAP

Differentiation

01 Differentiate

$$4x^3 + 2x$$

$$\frac{4}{x} = 4x^{-1}$$

$$\frac{5}{3x^4}$$

02 Gradient at a point, tangents

gradient is 4 $\rightarrow$ find x

tangent $y = ax + b$

find a and b

03 Turning points

$$\frac{dy}{dx} = 0$$

max or min?

negative gradient range

04 Maximum and minimum problems

cuboid volume

fence area

exact x

05 Kinematics

$s \rightarrow v \rightarrow a$

instantaneously at rest

minimum speed

THE ONE RULE Every question starts here:

ZONE RULE

$$y = ax^n$$

$$\frac{dy}{dx} = nax^{n-1}$$

ZONE MEANING

$$\frac{dy}{dx} = \text{gradient}$$

= 0 at a turning point

ZONE MOTION

$$s \rightarrow v \rightarrow a$$

differentiate once, then again

01 Differentiate

Write every term as a power of x first (fractions and roots too). Then for each term: multiply by the power, take one off. A number on its own disappears.

WHEN YOU SEE → YOU DO

$$y = 4x^3 + 2x$$

$$\rightarrow 12x^2 + 2$$

$$\frac{4}{x}, \frac{1}{3x}$$

$$\rightarrow 4x^{-1}, \frac{1}{3}x^{-1} \text{ first}$$

$$-\frac{5}{3x^4}$$

$$\rightarrow -\frac{5}{3}x^{-4} \rightarrow \frac{20}{3}x^{-5}$$

a constant (+7, -11)

→ its derivative is 0

Powers of x

× power

Power - 1

Tidy

WORKED MODEL • A FRACTION

Differentiate $y = \frac{2}{3}x^2 + \frac{1}{5}x - \frac{3}{2x}$.

1 $y = \frac{2}{3}x^2 + \frac{1}{5}x - \frac{3}{2}x^{-1}$

powers of x

ANSWER $\frac{dy}{dx} = \frac{4}{3}x + \frac{1}{5} + \frac{3}{2}x^{-2}$

× **TRAP** $\frac{4}{x}$ is not $4x$. Rewrite it as $4x^{-1}$ before you differentiate.

IDEA 01

Practise: Differentiate

My questions first, then selected practice.

Q1 **DR ESLAM** Powers first

CH 3 · TOPIC 10 2 marks

Differentiate $y = 3x^4 - \frac{6}{x} + 5$.**SELECTED PRACTICE****Q2** **PRACTICE** Practice · Medium

CH 3 · TOPIC 10 3 marks

1 (a) Use differentiation to find $\frac{dy}{dx}$ for the following:

$$y = x^4$$

(1 mark)

$$(b) y = 2x^{-3}$$

(1 mark)

$$(c) y = \frac{4}{x}$$

(1 mark)

Q3**PRACTICE**

Practice · Medium

CH 3 · TOPIC 10

3 marks

2 (a) Use differentiation to find $\frac{dy}{dx}$ for the following:

$$y = 4x^3 + 2x$$

(1 mark)

(b) $y = -5x^{-2}$

(1 mark)

(c) $y = \frac{1}{3x}$

(1 mark)

Q4**PRACTICE**

Practice · Medium

CH 3 · TOPIC 10 5 marks

3 (a) Use differentiation to find $\frac{dy}{dx}$ for the following:

$$y = 2x^3 - 6x^2 + 3x - 4$$

(1 mark)

(b) $-\frac{5}{3x^4}$

(2 marks)

(c) $\frac{2}{3}x^2 + \frac{1}{5}x - \frac{3}{2x}$

(2 marks)

02 Gradient at a Point and Tangents

The gradient at a point = $\frac{dy}{dx}$ at that x . If the gradient is given, set $\frac{dy}{dx}$ equal to it and solve for x .

WHEN YOU SEE → YOU DO

“the gradient at $x = -1.5$ ”

→ substitute into $\frac{dy}{dx}$

“the x coordinates where the gradient is 4”

→ $\frac{dy}{dx} = 4$, solve (a quadratic)

“the tangent at this point, $y = ax + b$ ”

→ $a = \text{gradient}$; point gives b

“(2, -6) lies on C and the gradient at A is 16”

→ two equations → a and b

WORKED MODEL • NOVEMBER 2025 P2H

$y = x^3 - 16x + 7$. The gradient is 11 at two points. Find the y -intercepts b of the tangents there.

1 $3x^2 - 16 = 11 \Rightarrow x = \pm 3$

gradient = 11

2 $x = 3: y = -14; x = -3: y = 34$

the points

ANSWER $b = -14 - 33 = -47$ or $b = 34 + 33 = 67$

IDEA 02

Practise: Gradient at a Point and Tangents

My questions first, then selected practice, then real past papers.

Q5 DR ESLAM Given gradient

CH 3 • TOPIC 10 3 marks

$y = x^3 - 3x^2 + 2$. Find the x coordinates where the gradient is 9.

SELECTED PRACTICE**Q6** PRACTICE Practice • Medium

CH 3 • TOPIC 10 4 marks

4 (a) For the curve with equation $y = 2x^2 - 6x - 11$:

find $\frac{dy}{dx}$

(2 marks)

(b) Find the coordinates of the point on the curve where the gradient is 2.

(2 marks)

Q7**PRACTICE**

Practice • Medium

CH 3 • TOPIC 10**7 marks**

5 (a) A curve has equation $y = x^3 + \frac{7}{2}x^2 - 2x + 9$

Find $\frac{dy}{dx}$

(2 marks)

(b) Find the gradient of the curve at the point where:

(i) $x = -3$

[2]

(ii) $x = \frac{2}{3}$

[2]

(4 marks)

(c) What can you say about the tangents to the curves at these two points?

(1 mark)

Q8**PRACTICE**

Practice · Medium

CH 3 · TOPIC 10 3 marks

10 Calculate the gradient of $y = 24 + 5x - x^2$ at $x = -1.5$.

(3 marks)

PAST PAPERS**Q9****PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q13

CH 3 · TOPIC 10 6 marks

13 The curve **C** has equation $y = 5x^3 - x^2 - 6x + 4$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

There are two points on the curve **C** at which the gradient of the curve is 2

(b) Find the x coordinate of each of these two points.
Show clear algebraic working.

.....

(Total for Question 13 is 6 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q24

CH 3 • TOPIC 10**6 marks**

24 The curve **C** has equation $y = ax^3 + bx^2 - 12x + 6$ where a and b are constants.

The point A with coordinates $(2, -6)$ lies on **C**

The gradient of the curve at A is 16

Find the y coordinate of the point on the curve whose x coordinate is 3

Show clear algebraic working.

$y = \dots\dots\dots$

(Total for Question 24 is 6 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q11

CH 3 • TOPIC 10**6 marks**

11 The curve **C** has equation $y = 4x^3 + x^2 - 20x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(b) Find the x coordinates of the points on **C** where the gradient is 4
Show clear algebraic working.

.....

(Total for Question 11 is 6 marks)

Q12**PAST PAPER**

Edexcel 4MA1 · November 2023 · Paper 1H · Q16

CH 3 · TOPIC 10**4 marks**

16 A curve has equation $y = 4x^3 - 8x + 5$

Find the x coordinates of the two points on the curve where the gradient is $\frac{1}{3}$

.....

(Total for Question 16 is 4 marks)

Q13**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1H · Q18

CH 3 · TOPIC 10**5 marks**

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

(Total for Question 18 is 5 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q20

CH 3 • TOPIC 10**4 marks**

20 The curve with equation $y = 2x^4 - 64x$ has a minimum point.

Find an equation of the tangent to the curve at the minimum point.

Show clear algebraic working.

.....
(Total for Question 20 is 4 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q17

CH 3 • TOPIC 10**6 marks**

17 $y = 4x^3 + 5x^2 + 2x$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots (2)$$

(b) Find the coordinates of the turning points on the graph with equation $y = 4x^3 + 5x^2 + 2x$
Show clear algebraic working.

.....
(Total for Question 17 is 6 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q22

CH 3 • TOPIC 10**6 marks**

22 Curve **C** has equation $y = x^3 - 16x + 7$

At two points on **C**, the gradient is 11

The tangents to **C** at these two points have equations of the form $y = ax + b$

Work out the two possible values of b

Show clear algebraic working.

(Total for Question 22 is 6 marks)

03 Turning Points

At a turning point the gradient is zero. Solve $\frac{dy}{dx} = 0$ for x , then substitute back for y . The shape tells you which is the maximum.

WHEN YOU SEE → YOU DO

“stationary / turning point”

→ $\frac{dy}{dx} = 0$

“find the coordinates”

→ x from $\frac{dy}{dx} = 0$, y from the curve

“which is a maximum?”

→ positive cubic: the left one is the max

“range of x where the gradient is negative”

→ $\frac{dy}{dx} < 0$: between the roots

dy/dx

Set = 0

Solve x

Substitute for y

WORKED MODEL • MAY 2024 P1HR

$y = x^3 - 8x^2 - 12x + 5$ has stationary points A and B, $x_A > x_B$. Find A.

1 $3x^2 - 16x - 12 = 0 \Rightarrow (3x + 2)(x - 6) = 0$

$\frac{dy}{dx} = 0$

2 $x_A = 6: 216 - 288 - 72 + 5$

substitute

ANSWER A(6, -139)

IDEA 03

Practise: Turning Points

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Turning points

CH 3 • TOPIC 10 4 marks

Find the turning points of $y = x^3 - 12x + 1$.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Medium

CH 3 • TOPIC 10

6 marks

8 (a) $y = x^3 - 6x^2 - 15x$.

Find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = \dots\dots\dots$$

(2 marks)

(b) The curve with equation $y = x^3 - 6x^2 - 15x$ has two stationary points.

Work out the coordinates of these two stationary points.

(4 marks)

Q19**PRACTICE**

Practice · Hard

CH 3 · TOPIC 10**8 marks**

1 (a) A curve has equation $y = 2x^2 + x - 3$.

Find: the coordinates where the curve crosses the x -axis,

(2 marks)

(b) the coordinates where the curve crosses the y -axis,

(1 mark)

(c) the coordinates of the turning point on the curve,

(3 marks)

(d) Sketch the curve showing the points you have found.

(2 marks)

Q20**PRACTICE**

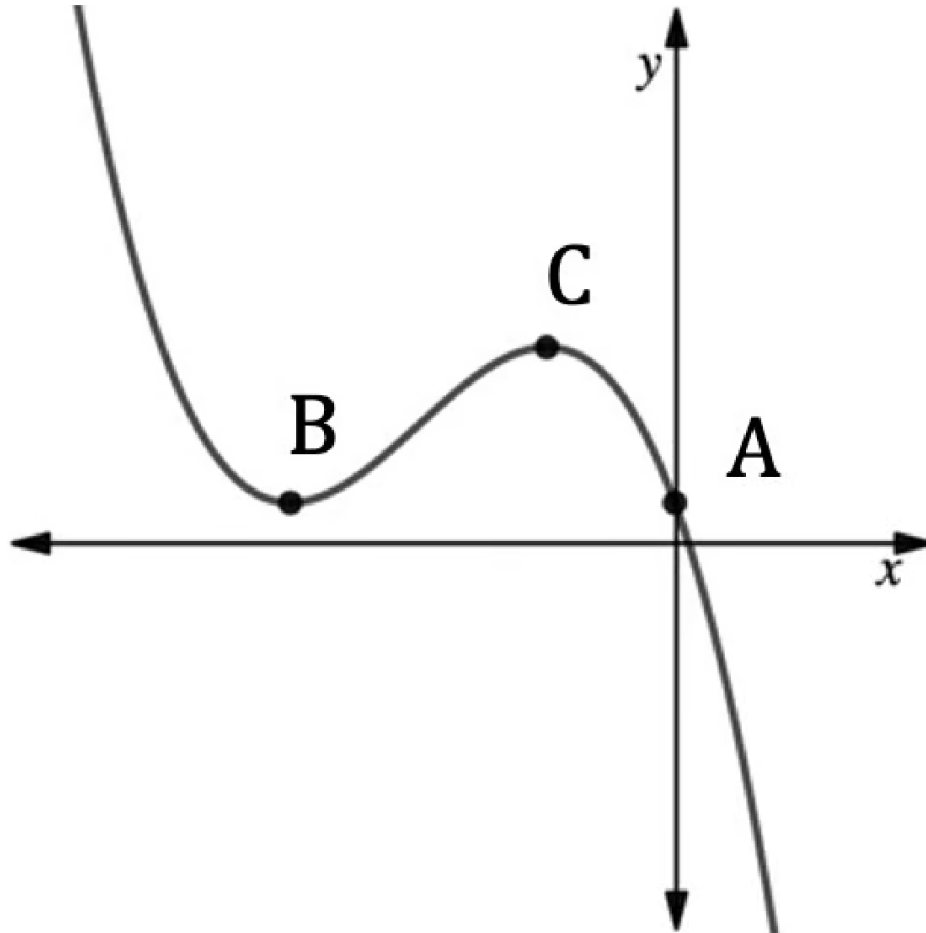
Practice · Hard

CH 3 · TOPIC 10

8 marks

- 4 (a) The curve G has equation $y = 1 - x^3 - 6x^2 - 9x$.

Part of the graph of G is shown below.



Write the coordinates of A.

(1 mark)

- (b) Points B and C are stationary points on G .

Find the coordinates of points B and C, stating the nature of the stationary point in each case.

(5 marks)

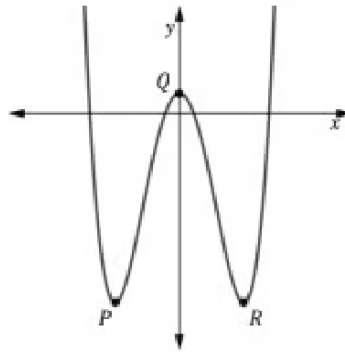
- (c) For which values of x is the gradient of the curve G negative?

(2 marks)

Q21 **PRACTICE** Practice · Very Hard

CH 3 · TOPIC 10 7 marks

- 2 Part of the graph with equation $y = 2x^4 - 16x^2 + 3$ is shown below.



The graph has three stationary points, indicated on the graph by points P , Q and R . Find the area of the triangle PQR .

(7 marks)

PAST PAPERS

Q22 **PAST PAPER** Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q23**CH 3 · TOPIC 10** **4 marks**

23 Curve C has equation $y = px^3 - mx$ where p and m are positive integers.

Find the range of values of x , in terms of p and m , for which the gradient of C is negative.

.....

(Total for Question 23 is 4 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q21

CH 3 • TOPIC 10 5 marks

- 21 The point A is the only stationary point on the curve with equation $y = kx^2 + \frac{16}{x}$ where k is a constant.

Given that the coordinates of A are $\left(\frac{2}{3}, a\right)$

find the value of a .

Show your working clearly.

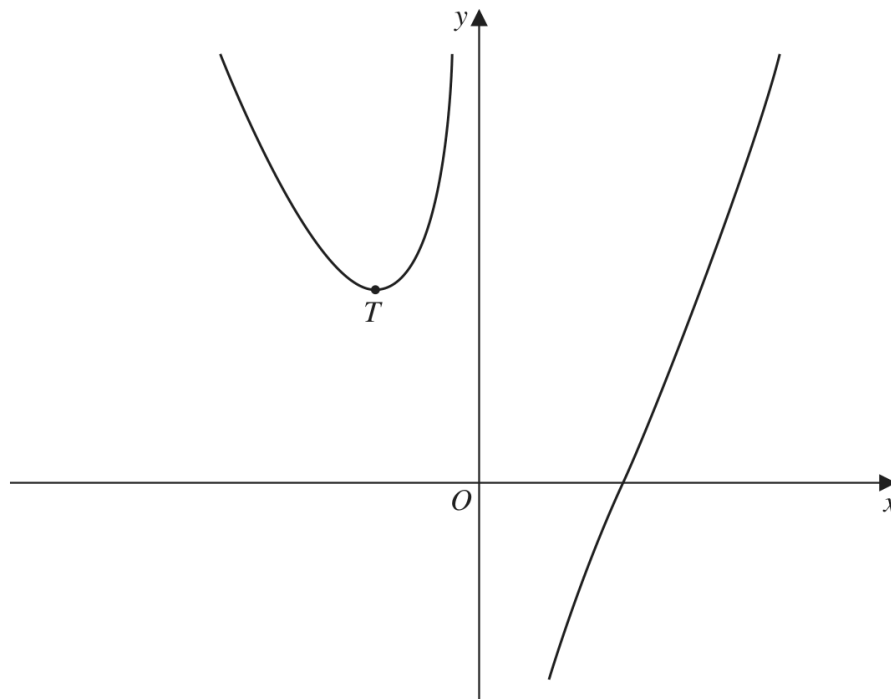
$a =$

(Total for Question 21 is 5 marks)

Q24 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1HR · Q22

CH 3 · TOPIC 10 7 marks

- 22 The diagram shows a sketch of part of the curve with equation $y = x^2 - \frac{p}{x}$ where p is a positive constant.

Diagram **NOT**
accurately drawn

For all values of p , the curve has exactly one turning point and this turning point is a minimum shown as the point T in the sketch.

For the curve where the x coordinate of T is -3

- (a) find the value of p

$$p = \dots\dots\dots$$

The line with equation $y = k$ is a tangent to the curve with equation $y = x^2 - \frac{16}{x}$

- (b) Find the value of k

$$k = \dots\dots\dots$$

(Total for Question 22 is 7 marks)

Q25**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q21

CH 3 • TOPIC 10**6 marks**

21 The curve **T** has equation $y = x^3 - 2x^2 - 9x + 15$

(a) Find $\frac{dy}{dx}$

$$\frac{dy}{dx} = \dots\dots\dots$$

(b) Find the range of values of x for which **T** has a positive gradient.
Give your values correct to 3 significant figures.
Show your working clearly.

.....
(Total for Question 21 is 6 marks)

Q26**PAST PAPER**

Edexcel 4MA1 · May 2024 · Paper 1HR · Q19

CH 3 · TOPIC 10**5 marks**

19 A curve C has equation $y = x^3 - 8x^2 - 12x + 5$

Curve C has exactly two stationary points, one at point A and one at point B such that

x coordinate of point $A > x$ coordinate of point B

Find the coordinates of point A

Show clear algebraic working.

(..... ,)

(Total for Question 19 is 5 marks)

04 Maximum and Minimum Problems

Write the quantity (volume, area) in one letter, differentiate, set to zero, solve. Then work out the maximum value itself.

WHEN YOU SEE → YOU DO

“show that $V = 15 + 16x - x^2 - 2x^3$ ”

→ length × width × height, expand

“the value of x that maximises V ”

→ $\frac{dV}{dx} = 0$, keep the positive root

“find the maximum volume”

→ substitute that x back into V

fence of fixed length / fixed total rods

→ use it to write l or y in terms of x

WORKED MODEL • JUNE 2025 P2HR

A cuboid is x by x by $(15 - 4x)$ cm. Find the maximum volume.

1 $V = 15x^2 - 4x^3$

volume

2 $30x - 12x^2 = 0 \Rightarrow x = 2.5$

$$\frac{dV}{dx} = 0$$

ANSWER $V = 15(6.25) - 4(15.625) = 31.25 \text{ cm}^3$

IDEA 04

Practise: Maximum and Minimum Problems

My questions first, then selected practice, then real past papers.

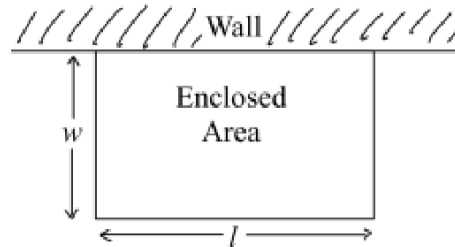
SELECTED PRACTICE

Q27**PRACTICE**

Practice · Hard

CH 3 • TOPIC 10 9 marks

- 7 (a) A homeowner wishes to enclose a rectangular part of their garden by building a fence, using an existing wall as one side of the rectangle as shown in the diagram below.



The width of the enclosed rectangle is w metres and its length l metres.

The homeowner has 40 metres of fence to use and would like to use it all in order to maximise the area of the garden to be enclosed. Show that $l = 40 - 2w$

(1 mark)

- (b) Show that the area of the garden to be enclosed, A , is given by $A = 40w - 2w^2$

(2 marks)

- (c) Find $\frac{dA}{dw}$

(2 marks)

- (d) Find the value of w that maximises A

(2 marks)

- (e) Find the dimensions of the rectangle that produce the maximum area that can be enclosed using all of the fence.

Also find the maximum area.

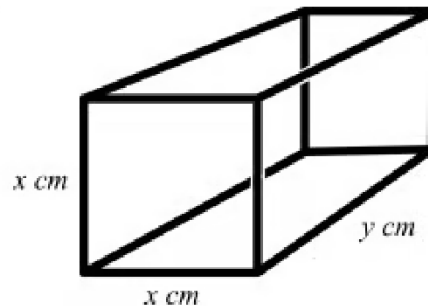
(2 marks)

Q28**PRACTICE**

Practice • Hard

CH 3 • TOPIC 10 11 marks

- 10 (a)** A cuboid with a square cross section is to be made from rods as shown in the diagram. The shorter rods making the square are of length x cm and the longer rods



are of length y cm.

Explain why 12 rods in total will be needed to make the cuboid, and state how many of each length will be required.

The total length of the rods is to be fixed at 36 cm.

(2 marks)

- (b)** The total length of the rods is to be fixed at 36 cm.

Find y in terms of x

(2 marks)

- (c)** Show that the volume of the cuboid, $V \text{ cm}^3$ is $V = 9x^2 - 2x^3$.

(2 marks)

- (d)** Find the value of x that maximises the volume.

(3 marks)

- (e)** Find the maximum volume.

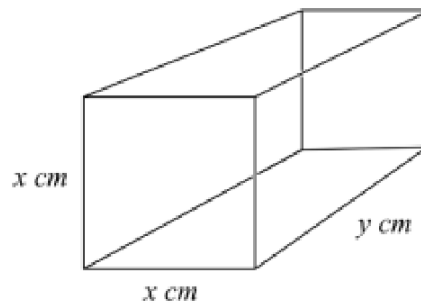
(2 marks)

Q29**PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 10 10 marks

- 3 (a) The diagram shows a cuboid with a square cross-section.



The sides of the square face are x cm and the length of the cuboid is y cm.
The cuboid is to have a fixed surface area, A , of 25 cm^2 .

Show that the volume of the cuboid, $V \text{ cm}^3$ is given by

$$V = \frac{25}{4}x - \frac{1}{2}x^3$$

(4 marks)

- (b) Show that the value of x that maximises the volume of the cuboid is $\frac{5\sqrt{6}}{6}$

(4 marks)

- (c) Find the maximum volume of the cuboid, correct to 3 significant figures.

(2 marks)

Q30

PRACTICE

Practice • Very Hard

CH 3 • TOPIC 10 6 marks

9 (a) $ABCED$ is a five-sided shape.

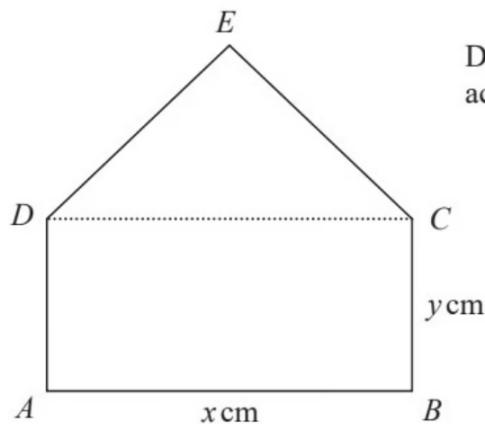


Diagram **NOT**
accurately drawn

$ABCD$ is a rectangle.

CED is an equilateral triangle.

$$AB = x \text{ cm} \quad BC = y \text{ cm}$$

The perimeter of $ABCED$ is 100 cm.

The area of $ABCED$ is $R \text{ cm}^2$

$$\text{Show that } R = \frac{x}{4}(200 - [6 - \sqrt{3}]x)$$

(3 marks)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$$x = \dots\dots\dots [2]$$

(ii) Explain why the maximum value of R is given by this value of x .

[1]

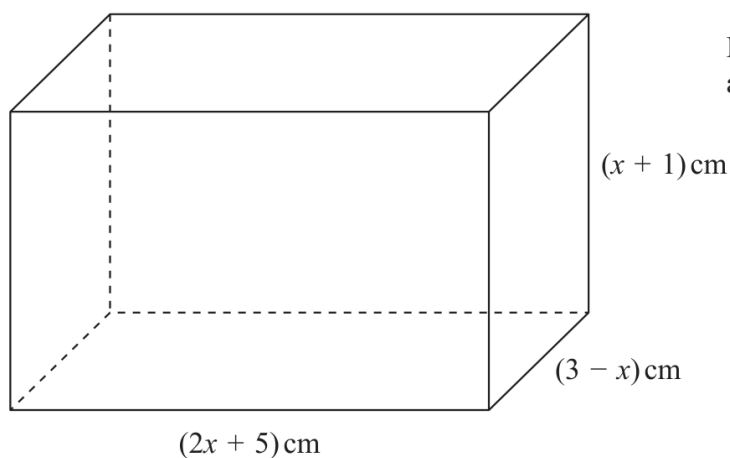
(3 marks)

PAST PAPERS

Q31 PAST PAPER Edexcel 4MA1 • January 2020 • Paper 1H • Q15

CH 3 • TOPIC 10 8 marks

15

Diagram **NOT**
accurately drawn

The diagram shows a cuboid of volume $V\text{cm}^3$

(a) Show that $V = 15 + 16x - x^2 - 2x^3$

There is a value of x for which the volume of the cuboid is a maximum.

(b) Find this value of x .

Show your working clearly.

Give your answer correct to 3 significant figures.

$x = \dots\dots\dots$

(Total for Question 15 is 8 marks)

Q32**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1H · Q17

CH 3 · TOPIC 10 5 marks

17 A solid, **S**, is made from a hemisphere and a cylinder.

The centre of the circular face of the hemisphere and the centre of the top face of the cylinder are at the same point.

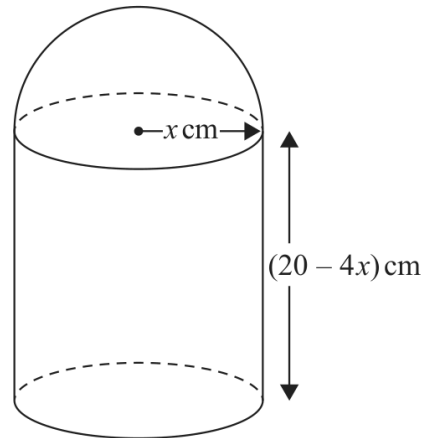


Diagram **NOT**
accurately drawn

The radius of the cylinder and the radius of the hemisphere are both x cm.
The height of the cylinder is $(20 - 4x)$ cm.

The volume of **S** is $V \text{ cm}^3$ where $V = \frac{1}{3} \pi y$

Find the maximum value of y .
Show clear algebraic working.

(Total for Question 17 is 5 marks)

Q33**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2HR • Q22

CH 3 • TOPIC 10**5 marks**

22 The diagram shows a solid cuboid.

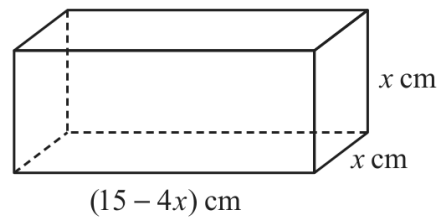


Diagram **NOT**
accurately drawn

The volume of the cuboid is $V \text{ cm}^3$

Find the maximum value of V

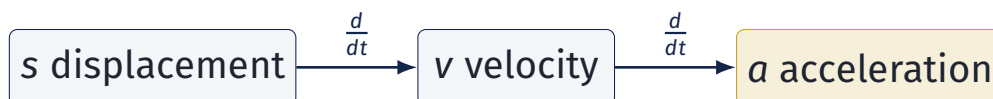
.....
(Total for Question 22 is 5 marks)

05 Kinematics: s, v, a

Displacement $\rightarrow$ differentiate $\rightarrow$ velocity $\rightarrow$ differentiate $\rightarrow$ acceleration. “At rest” means $v = 0$.

WHEN YOU SEE $\rightarrow$ YOU DO

“instantaneously at rest”	$\rightarrow v = \frac{ds}{dt} = 0$
“acceleration when ...”	$\rightarrow a = \frac{dv}{dt}$
“minimum speed”	$\rightarrow a = 0$, then find v
“direction reverses”	$\rightarrow v = 0$ (and changes sign)
“moving in the same direction”	$\rightarrow$ both velocities have the same sign



WORKED MODEL • JANUARY 2021 P2H

$s = t^3 + 5t^2 - 8t + 10$. Find the displacement when P is instantaneously at rest.

1 $v = 3t^2 + 10t - 8 = 0 \Rightarrow (3t - 2)(t + 4) = 0$

$v = 0$

2 $t = \frac{2}{3} \ (t \geq 0)$

keep the valid time

ANSWER $s = \frac{8}{27} + \frac{20}{9} - \frac{16}{3} + 10 = \frac{194}{27} \text{ m}$

IDEA 05

Practise: Kinematics: s, v, a

My questions first, then selected practice, then real past papers.

Q34**DR ESLAM**

At rest

CH 3 • TOPIC 10

3 marks

$s = t^3 - 6t^2 + 9t$. Find the times when the particle is at rest.

SELECTED PRACTICE

Q35**PRACTICE**

Practice • Medium

CH 3 • TOPIC 10 6 marks

- 6 (a) A particle P passes the fixed point O whilst moving along a straight line.

The displacement of P , from O , at time t seconds is s metres where

$$s = 6t^3 - 12t^2 + 7t$$

Find expressions for the velocity, $v \text{ m/s}$, and the acceleration, $a \text{ m/s}^2$ of the particle at time t seconds.

(4 marks)

- (b) Find the time at which the acceleration is 3 m/s^2 .

(2 marks)

Q36**PRACTICE**

Practice · Hard

CH 3 · TOPIC 10**4 marks**

2 (a) A particle is moving along a straight line. The fixed point O lies on this line.

The displacement of the particle from O at time t seconds is s metres where

$$s = 2t^3 - 9t^2 - 60t$$

Find an expression for the velocity, v m/s of the particle at time t seconds.

(2 marks)

(b) Find the time at which the velocity is instantaneously zero.

(2 marks)

Q37**PRACTICE**

Practice • Hard

CH 3 • TOPIC 10 7 marks

6 (a) A particle is moving along a straight line and passes a fixed point O .

The displacement of the particle, from point O , at time t seconds is

$$s = \frac{1}{3}t^3 - \frac{5}{2}t^2 + 20t - 15$$

where s is measured in metres. Initially how far is the particle from O ?

(1 mark)

(b) Find, in terms of t , the velocity of the particle.

(2 marks)

(c) Find the time at which the particle's velocity is at its minimum.

(2 marks)

(d) For how long is the particle decelerating?

(2 marks)

PAST PAPERS

Q38**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1HR • Q23

CH 3 • TOPIC 10**6 marks**

23 A particle moves along a straight line.

The fixed point O lies on this line.

The displacement of the particle from O at time t seconds, $t \geq 0$, is s metres where

$$s = t^3 + 4t^2 - 5t + 7$$

At time T seconds the velocity of P is V m/s where $V \geq -5$

Find an expression for T in terms of V .

Give your expression in the form $\frac{-4 + \sqrt{k + mV}}{3}$ where k and m are integers to be found.

$T = \dots\dots\dots$

(Total for Question 23 is 6 marks)

Q39**PAST PAPER**

Edexcel 4MA1 · January 2021 · Paper 1HR · Q20

CH 3 · TOPIC 10 5 marks

- 20** A particle P is moving along a straight line.
The fixed point O lies on the line.

At time t seconds ($t \geq 0$), the displacement of P from O is s metres where

$$s = t^3 - 9t^2 + 33t - 6$$

Find the minimum speed of P .

..... m/s

(Total for Question 20 is 5 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2H • Q19

CH 3 • TOPIC 10**5 marks**

19 A particle P is moving along a straight line.

The fixed point O lies on this line.

At time t seconds where $t \geq 0$, the displacement, s metres, of P from O is given by

$$s = t^3 + 5t^2 - 8t + 10$$

Find the displacement of P from O when P is instantaneously at rest.

Give your answer in the form $\frac{a}{b}$ where a and b are integers.

..... metres

(Total for Question 19 is 5 marks)

Q41 **PAST PAPER** Edexcel 4MA1 • January 2022 • Paper 1HR • Q23**CH 3 • TOPIC 10** **6 marks**

- 23** Two particles, P and Q , move along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds is s metres, where

$$s = t^3 - 4t^2 + 5t \quad \text{for } t > 1$$

The displacement of Q from O at time t seconds is x metres, where

$$x = t^2 - 4t + 4 \quad \text{for } t > 1$$

Find the range of values of t where $t > 1$ for which both particles are moving in the same direction along the straight line.

.....
(Total for Question 23 is 6 marks)

Q42**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q24

CH 3 • TOPIC 10**5 marks**

24 A particle P moves along a straight line that passes through the fixed point O

The displacement, x metres, of P from O at time t seconds, where $t \geq 0$, is given by

$$x = 4t^3 - 27t + 8$$

The direction of motion of P reverses when P is at the point A on the line.

The acceleration of P at the instant when P is at A is $a \text{ m/s}^2$

Find the value of a

$a = \dots\dots\dots$

(Total for Question 24 is 5 marks)

Q43**PAST PAPER**

Edexcel 4MA1 · June 2022 · Paper 1H · Q17

CH 3 · TOPIC 10 5 marks

- 17 A particle P moves along a straight line.
The fixed point O lies on this line.

The displacement of P from O at time t seconds, $t \geq 1$, is s metres where

$$s = 4t^2 + \frac{125}{t}$$

The velocity of P at time t seconds, $t \geq 1$, is v m/s

Work out the distance of P from O at the instant when $v = 0$

..... m

(Total for Question 17 is 5 marks)

Q44**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q18

CH 3 • TOPIC 10**4 marks**

- 18 A particle is moving along a straight line that passes through the fixed point O
The displacement, s metres, of the particle from O at time t seconds is given by

$$s = 2t^3 - 5t^2 + 6t - 5$$

Find the value of t when the acceleration of the particle is 5 m/s^2

$t = \dots\dots\dots$

(Total for Question 18 is 4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|---|
| 01 | Differentiate | Powers of x first; $ax^n \rightarrow nax^{n-1}$. |
| 02 | Gradient | Substitute x ; or set $\frac{dy}{dx} = \text{given value}$ and solve. |
| 03 | Turning points | $\frac{dy}{dx} = 0$, then y . |
| 04 | Max / min | One letter, differentiate, $= 0$, substitute back. |
| 05 | Motion | $s \rightarrow v \rightarrow a$; at rest: $v = 0$. |

BOTTOM LINE Power down, take one off $\rightarrow$ derivative = gradient $\rightarrow$ zero at the top and bottom.

Answers

Differentiation • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • DIFFERENTIATE

- | | | |
|-----------|-------------------------|--|
| Q1 | Dr Eslam • Powers first | $12x^3 + 6x^{-2}$ |
| Q2 | Practice • Medium | (a) $4x^3$ (b) $-6x^{-4}$ (c) $-4x^{-2}$ |
| Q3 | Practice • Medium | (a) $12x^2 + 2$ (b) $10x^{-3}$ (c) $-\frac{1}{3}x^{-2}$ |
| Q4 | Practice • Medium | (a) $6x^2 - 12x + 3$ (b) $\frac{20}{3}x^{-5}$ (c) $\frac{4}{3}x + \frac{1}{5} + \frac{3}{2}x^{-2}$ |

IDEA 02 • GRADIENT AT A POINT AND TANGENTS

- Q5** Dr Eslam • Given gradient $3x^2 - 6x - 9 = 0$: $x = 3$ or -1
- Q6** Practice • Medium (a) $4x - 6$ (b) $(2, -15)$
- Q7** Practice • Medium (a) $3x^2 + 7x - 2$ (b)(i) 4 (ii) 4 (c) they are parallel
- Q8** Practice • Medium 8
- Q9** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q13 (a) $15x^2 - 2x - 6$ (b) $x = \frac{4}{5}$ or $-\frac{2}{3}$
- Q10** Edexcel 4MA1 • January 2022 • Paper 1H • Q24 $a = 4, b = -5$: $y = 33$
- Q11** Edexcel 4MA1 • May 2023 • Paper 1H • Q11 (a) $12x^2 + 2x - 20$ (b) $x = \frac{4}{3}$ or $-\frac{3}{2}$
- Q12** Edexcel 4MA1 • November 2023 • Paper 1H • Q16 $x = \pm\frac{5}{6}$
- Q13** Edexcel 4MA1 • May 2024 • Paper 1H • Q18 $(4, -95)$ and $(-4, 97)$
- Q14** Edexcel 4MA1 • November 2024 • Paper 1H • Q20 min $(2, -96)$: $y = -96$
- Q15** Edexcel 4MA1 • June 2025 • Paper 2H • Q17 (a) $12x^2 + 10x + 2$ (b) $\left(-\frac{1}{2}, -\frac{1}{4}\right), \left(-\frac{1}{3}, -\frac{7}{27}\right)$
- Q16** Edexcel 4MA1 • November 2025 • Paper 2H • Q22 $b = -47$ or 67

IDEA 03 • TURNING POINTS

- Q17** Dr Eslam • Turning points $3x^2 - 12 = 0$: (2, -15) and (-2, 17)
- Q18** Practice • Medium (a) $3x^2 - 12x - 15$ (b) (-1, 8) and (5, -100)
- Q19** Practice • Hard (a) (-1.5, 0), (1, 0) (b) (0, -3) (c) $\left(-\frac{1}{4}, -\frac{25}{8}\right)$
- Q20** Practice • Hard (a) (0, 1) (b) B(-3, 1) minimum, C(-1, 5) maximum (c) $x < -3$ or $x > -1$
- Q21** Practice • Very Hard P(-2, -29), Q(0, 3), R(2, -29): area 64
- Q22** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q23 $-\sqrt{\frac{m}{3p}} < x < \sqrt{\frac{m}{3p}}$
- Q23** Edexcel 4MA1 • May 2021 • Paper 2H • Q21 $k = 18$: $a = 36$
- Q24** Edexcel 4MA1 • June 2022 • Paper 1HR • Q22 (a) $p = 54$ (b) $k = 12$
- Q25** Edexcel 4MA1 • May 2023 • Paper 1HR • Q21 (a) $3x^2 - 4x - 9$ (b) $x < -1.19$ or $x > 2.52$
- Q26** Edexcel 4MA1 • May 2024 • Paper 1HR • Q19 A(6, -139)

IDEA 04 • MAXIMUM AND MINIMUM PROBLEMS

- Q27** Practice • Hard (c) $40 - 4w$ (d) $w = 10$ (e) 10 m by 20 m, 200 m²
- Q28** Practice • Hard (b) $y = 9 - 2x$ (d) $x = 3$ (e) 27 cm³
- Q29** Practice • Very Hard (c) 8.51 cm³
- Q30** Practice • Very Hard (b)(i) $x = \frac{100}{6 - \sqrt{3}}$ (ii) R is a quadratic in x
- Q31** Edexcel 4MA1 • January 2020 • Paper 1H • Q15 (b) $x = 1.47$
- Q32** Edexcel 4MA1 • January 2021 • Paper 1H • Q17 $x = 5: y = 320$
- Q33** Edexcel 4MA1 • June 2025 • Paper 2HR • Q22 $x = 2.5: V = 31.25 \text{ cm}^3$

IDEA 05 • KINEMATICS: s, v, a

- Q34** Dr Eslam • At rest $3t^2 - 12t + 9 = 0: t = 1, t = 3$
- Q35** Practice • Medium (a) $v = 18t^2 - 24t + 7, a = 36t - 24$ (b) $t = \frac{3}{4}$
- Q36** Practice • Hard (a) $6t^2 - 18t - 60$ (b) $t = 5$
- Q37** Practice • Hard (a) 15 m (b) $t^2 - 5t + 20$ (c) $t = 2.5$ (d) for 2.5 s
- Q38** Edexcel 4MA1 • January 2020 • Paper 1HR • Q23 $T = \frac{-4 + \sqrt{31 + 3V}}{3}: k = 31, m = 3$
- Q39** Edexcel 4MA1 • January 2021 • Paper 1HR • Q20 6 m/s
- Q40** Edexcel 4MA1 • January 2021 • Paper 2H • Q19 $\frac{194}{27} \text{ m}$
- Q41** Edexcel 4MA1 • January 2022 • Paper 1HR • Q23 $1 < t < \frac{5}{3} \text{ or } t > 2$
- Q42** Edexcel 4MA1 • January 2022 • Paper 2H • Q24 $a = 36$
- Q43** Edexcel 4MA1 • June 2022 • Paper 1H • Q17 75 m
- Q44** Edexcel 4MA1 • November 2023 • Paper 2H • Q18 $t = \frac{5}{4}$